

JK 12-16 Python First Programmer
Lesson 03 : Input & Outputs

Duration : 90 mins

Description

In this lesson, students will learn how to give input to the programs that they will create by also incorporating some modules from Python library such as random module to generate random number and shutil module to create an archive file. Furthermore, students will also learn how to combine input and output in a project to get dynamic results.

Objectives

Students will be able to:

1. Understand what a module is and how to use it in a python code.
2. Show what random module is and apply it to generate random number.
3. Describe how they can use input() function to give input to the program.
4. Demonstrate how to use input() function combined with outputs.
5. Create various projects incorporating input and output concepts.

Tools

1. Pycharm

Class Structure

Duration	Topic
5 mins	Preparation
5 mins	Comprehension Check
15 mins	Random Number
15 mins	Input & Outputs
10 mins	Make Archive
10 mins	Simple Story
15 mins	Magical Calculator
10 mins	Review Quiz
5 mins	Wrap Up

Planned Activities	Resources
Preparation (5 mins)	
- Teacher greets students and ask how's their conditions. - Instruct students to open their pycharm. - If possible, teacher can incorporate a mini game during this preparation time to engage with the students.	
Comprehension Check (5 mins)	
- Teacher ask students if they are still remember about the materials from the previous lesson. - Teacher check on students' comprehension regarding the previous lesson by asking relatable questions. - Teacher explain and demonstrate in brief or summarize about what they have learned from the previous lesson.	
Random Number (15 mins)	
Explain – Random Number - We explain to students that for today, we will create a project that requires us to randomize the number inside our variable. - Therefore, first, we need to understand how to use the random function.	File/Code: 1.RandomNumber.py

- We ask students to create a new project and show them the random number project. We explain to students that every time we run the project, the number displayed will always be different.

Explain and Show – Random Function

- We then explain that to use the random function, we will need to import it first.
- We can explain that the random function is actually a really complex function – but because there are people that have created those functions, us users can then use it by importing it to our project.
- Next, after we have imported the function, we can then use it. To use it, we can type **random.randrange(x , y)** where inside x and y we can specify the range of random numbers that we want.

Explain and Show – Program Structure

- We explain that when we create our program, it is really important for us to create a program with a tidy program structure.
- For example, in the example, we were able to divide the different parts and add appropriate comments to better explain our code.

Individual Activity – Project Creation

- We ask students to add the random function to their project and create a simple program where when we run it, it will display the random number.
- We also ask students so that when they create the project, they should structure it properly and add appropriate comments.

Input & Outputs (15 mins)

Explain and Code – Input Asking

- We explain to students that for the next step, we would like to create a project where we will ask the user for the answer.
- To do this, we need input from the user.
- To add input, we show students the project example and explain to students that they can add it by using the input function.
- This input function will then display the message in the quotation mark and store the answer inside the variable.
- We ask students to try the input function and ask them to display the input.

File/Code:

[2.InputOutput.py](#)

Explain and Show – Simple Guess The Number

- We then show students the next project that they should create and explain to them that inside the project, we will ask the user to input a number between one to three.
- After the user answers, the user will then be told that the answer is right or wrong through a “True” or “False” message.

File/Code:

[3.GuessTheNumber.py](#)

Individual Activity – Creating The Project

- We ask students about how they think the project is created and tell them that for the project, they should use comparison operators.
- We tell students to do it by themselves first and if they have some difficulties, we can tell them that they can work with the person beside them and discuss it together.

Make Archive (10 mins)

Explain and Show – Mini Project

- We show students mini projects and explain to them how to practice their programming; they will make projects, make popular programs, **packaging files into zip**.

File/Code:

[4.MiniProject.py](#)

- We explain to students that this program will collect and bundle some files into a zip. We explain the name of the zip file will be randomized using random function.

Individual Activity – Mini Project Creation

- We tell students that now that they know how the project might work, they should try and create it on their own.
- We remind students that in order to create the program successfully, they have to use some of the skills that they have learned during the lesson.
- We give students time to do it by themselves and discuss it together when they are finished.

Simple Story (10 mins)

Explain and Show – Simple Story Project

- We show students the project and explain that for their next task, we would like them to create a project that focuses on input and output.
- For example, inside the simplestory.py project we tried to create a simple story by prompting the user to input several data such as names, hobbies, and friends.
- These data that was filled out by the user are then immediately displayed on the screen in the form of a short story.

File/Code:

[5.SimpleStory.py](#)

Individual Activity – SimpleStory Project

- We tell students that for the next project, they should create their own simple story project and incorporate at least 4 input prompts for the user.
- We can tell students to create the project and that they should try to use their own creativity when creating the project.

Group Activity – Trying Out Project

- When students are finished, we can tell students to switch seats and try the stories that were created by other students.
- We can then ask students which stories did they find the most interesting and why?

Magical Calculator (15 mins)

Explain and Show – Magical Calculator

- We show students the project and explain that for today's last task, they will have to create a program that will calculate the values that were inputted by the users.
- We analyze the project together and ask students what they think we will need for the project.
- We then explain that we will need to add arithmetic operators as a way for us to calculate the numbers.

File/Code:

[6.MagicalCalculator.py](#)

Individual Activity – Magical Calculator

- We tell students to create the project by themselves and that if they have any difficulties, they can ask us for assistance.
- During their project creation, students might be confused with how their divided number is not in decimal form. When this occurs or when they are nearly finished with the project, we can explain that in order to incorporate true division, we can import a function that will enable us to do so.
- The import statement is as follow: **from __future__ import division.**
- We give students time and discuss the project together when they are finished.

Review Quiz (10 mins)

Individual Activity – Review Quiz

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| <ul style="list-style-type: none">• We show students the review quiz questions for today and give them some time to answer it by themselves in a notepad file.• When everyone is done, we tell them to save their answers and review it together. | |
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Wrap Up (5 mins)	
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Wrapping Up	
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| <ul style="list-style-type: none">• We sum up by asking questions, making sure the students remember the most important things.• We thank students for doing a great job overall. | |
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☐ ☆ Good Luck With The Class !! ☆ ☐